

11.4 Storage

James can be used in conjunction with RDBMS tools to store mails and other data. It connects to databases through customised JDBC (Java Database Connectivity) drivers. James is particularly well-integrated with MySQL, a free and open source server. The driver for MySQL is packaged with the installation binaries. For other databases, download the appropriate driver into the `lib` folder of the James installation.

Browse to the `<database-connections>` block in `config.xml` and configure the children in the `<data-source>` element. There are a number of data-source elements each with a unique attribute name. The children in each data-source element are driver (class name of database driver), dburl (JDBC connection URL to the database driver), user, password, and max (maximum number of concurrent JDBC connections). A sample configuration for MySQL is shown below.

```
<data-source name="maildb"
  class="org.apache.james.util.mordred.
  JdbcDataSource">
  <driver> org.gjt.mm.mysql.Driver </driver>
  <dburl> jdbc:mysql://1211.0.0.1/mail </dburl>
  <user> username </user>
  <password> password </password>
  <max> 20 </max>
</data-source>
```

Further modifications are needed in the `config.xml` file for different data types like mails and message spools to be stored in separate locations. For example, to store mails in the maildb data-source, the following changes to the `<inboxRepository>` tag are needed:

```
<inboxRepository>
  <repository destinationURL="db://maildb/inbox/"
```